NAM HOUNG - 1 HYDRO POWER PROJECT

Document Prepared by Kosher Climate India Private Limited

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Project Title	Nam Houng - 1 Hydro Power Project
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CONTENTS

1 PROJECT DETAILS

1.1 Summary Description of the Project

Nam Houng - 1 Hydro Power Project (hereafter referred to as the “the project”) is developed by Nam Houng - 1 Hydro Power Co., Ltd in Xayabouly province, Lao PDR. The primary purpose of the project is to generate electricity to the Lao Power Grid.

The project activity is a hydro power project with a total installed capacity of 15 MW. The project is promoted by Nam Houng - 1 Hydro Power Co., Ltd. Project was commissioned on 01-September-2023 with a net annual generation of 87,280 MWh.

The project is expected to constantly contribute clean energy to the Lao Power Grid. It not only conforms to the present situation, but also can adapt to future changes in the electricity market. The baseline scenario of the project is continuation of the present situation, i.e. electricity supplied from the power grid. By displacing part of the power generated by thermal power plants, the project is therefore expected to reduce CO_2 emissions by an estimation of 31,453 tCO_2e per year during the first crediting period of 7 years.

The project is not only supplying renewable electricity to grid, but also producing positive environmental and economic benefits along with the promotion of the local sustainable development in the following aspects:

- During the construction period, plenty of job opportunities were provided to local residents,
- The infrastructure has been greatly improved. The enhancement of the transportation and electricity system brings substantial benefits to local villagers;
- The use of firewood will be reduced because of displacement by electricity, this reduces the damage to the local vegetation;
- Provide clean & cheap electricity in this region, promote the sustainable development in this region and slowing down the increasing trend of GHG emissions.

The Nam Houng - 1 Hydro Power development project is a part of the power development strategy by the Xayabouly Provincial Government Lao PDR to achieve the following important objectives:

- Augmenting foreign exchange earnings for Lao PDR by selling surplus electrical energy to consumption in the northern grid to Thailand.
- Assisting in the displacement of imported kerosene and diesel fuels by indigenous hydropower thereby reducing the outflow of foreign exchange.
- Assisting in the reduction of imported power from neighbor countries.
- Assisting in the reduction of Energy losses and stability on the EDL grid.
- Facilitating the development of agro-based and other industries.
- Improving the general standard of living of the population of Lao PDR.

- One part in Xayabouly Province has to be purchased energy by Lao Government Policy in year 2020 on 90% electrified whole country.

Audit Type	Period	Program	VVB Name	Number of years
Validation	13-December-2023 to 29-February 2024	VCS	4K Earth Science Private Limited	-
Total	13-December-2023 to 29-February 2024	VCS	4K Earth Science Private Limited	-

1.2 Sectoral Scope and Project Type

Sectoral Scope : 01 – Energy Industries (Renewable/non-renewable sources)
Project Type : Renewable Energy Project

The project is not AFOLU project and is not a grouped project.

1.3 Project Eligibility

The project is a 15 MW small-scale grid connected electricity generation project using hydro power in Lao PDR which is a Least developed Country (LDC)[1] as per the UN. As per VCS standard V-4.5, section 2.1.3 – Table 1, the small-scale hydro power project activity in LDC does not fall under excluded project activities. Hence, the project activity is eligible under VCS.

1.4 Project Design

☒ The project includes a single location or installation only

☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project

☐ The project is a grouped project

The project has been designed to include a single installation of an activity.

Eligibility Criteria

The project is not a grouped project. Thus, this section is not applicable.

[1] https://unctad.org/topic/least-developed-countries/list

1.5 Project Proponent

Organization name	Nam Houng - 1 Hydro Power Co., Ltd
Contact person	Phapithack Ekaphanh
Title	Director
Address	Khampheng Mueang Road (T4), Phonthan Nuea Village, Sayysathar District, Vientiane Capital
Telephone	+85621 922777
Email	smgroad@yahoo.com info@smgroup.com.la

1.6 Other Entities Involved in the Project

Organization name	Kosher Climate India Private Limited
Role in the project	Project Representative
Contact person	Vamsi Krishna M
Title	Director
Address	Zee Plaza, No.1678, 27th Main Rd, Near KLM Mall, Sector 2, HSR Layout, Bengaluru, Karnataka 560102, India
Telephone	+91 99453 43475
Email	vamsi@kosherclimate.com narendra@kosherclimate.com

1.7 Ownership

Nam Houng - 1 Hydro Power Co., Ltd. is a SPV of Simoung Group Co., Ltd. Simoung Group Co., Ltd. is the legal owner of Nam Houng – 1 Hydro Power Project (Nam Houng - 1 Hydro Power Co., Ltd.). They have the legal right to control and operate the project activities.

The business license of the project owner are the evidences for legislative right. Besides, the project allotment license, the equipment purchase contract and the power purchase agreement are the evidences for the ownership of the plant equipment and power generation.

1.8 Project Start Date

01-September-2023

1.9 Project Crediting Period

Project Owner has chosen the renewable crediting period with first crediting of 7 Years which starts from 01-September-2023 (Commissioning date) to 31-August-2030 with an option of two times of renewable up to 21 years.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

☐ <20,000 tCO_2e/year

☒ 20,000 – 100,000 tCO_2e/year

☐ 100,001 – 1,000,000 tCO_2e/year

☐ >1,000,000 tCO_2e/year

Project Scale	
Project	X
Large project	

Year	Estimated GHG emission reductions or removals (tCO_2e)
01-Septmeber-2023 to 31-December-2023	10,513
01-January-2024 to 31-December-2024	31,453
01-January-2025 to 31- December-2025	31,453
01-January-2026 to 31- December-2026	31,453
01-January-2027 to 31- December-2027	31,453
01-January-2028 to 31- December-2028	31,453
01-January-2029 to 31- December-2029	31,453
01-January-2030 to 31-August-2030	20,940
Total estimated ERs	**220,174**
Total number of crediting years	**7**
Average annual ERs	**31,453**

1.11 Description of the Project Activity

The project activity is a Hydro power project with a total installed capacity of 15 MW. The project is promoted by Nam Houng - 1 Hydro Power Co., Ltd which is a SPV of Simoung Group Co., Ltd. Simoung Group Co., Ltd. is the legal owner of Nam Houng-1 hydro power project.

The Project activity is a new facility (Greenfield) and the electricity generated by the project will be exported to the electricity grid. The project will therefore displace an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid. The Project Proponent plans to avail the VCS benefits for the project.

In the Pre- project scenario, the entire electricity delivered to the grid by the project activity, would have otherwise been generated by the operation of grid-connected power plant. The scenario prior to the start of implementation of the project activity is generation of electricity, dominated by thermal power plants, thus leading to GHG emissions. The baseline scenario is the same as the scenario prior to the start of implementation of the project activity. The project shall result in replacing anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 31,453 tCO_2e per year, thereon displacing 87,280 MWh/year amount of electricity from the gird over the 7 years crediting period. The hydro project structure consists of release sluice, main powerhouse, and right-bank non overflow dam, etc.

Table 1: Main Technical Parameters of the project

Turbine		
Item	Value	Unit
Number of units	3	set
Rated Capacity	5	MW
Maximum working head	22	m
Least working head	14.67	m
Rated Head	21	m
Rated speed	250	rpm
Rated flow	28.7	m^3/s
Turbine type	Bulb	
Lifetime	30	years
Generator		
Item	Value	Unit
Number of units	3	set
Rated Capacity	5	MW
Generator Rated Power Factor	0.85	
Generator efficiency	97	%
Lifetime	30	Years

1.12 Project Location

The project is located in the Xayabouly Province, Lao PDR.

Table 2: Project Location

Project Location	Latitude	Longitude
Xayabouly Province, Lao PDR	19°11'19.7"N	101°48'17.8"E

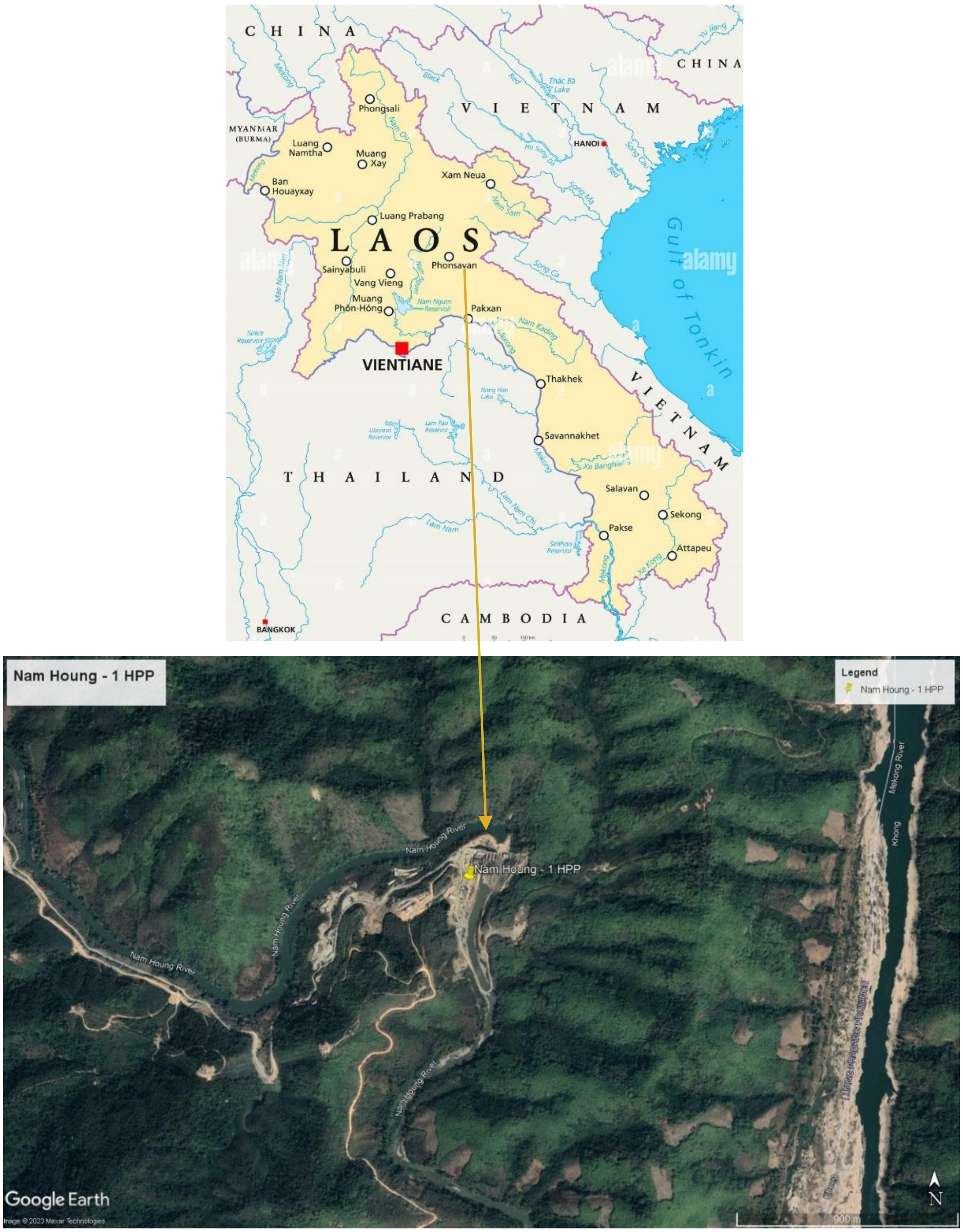

Fig.1 Project Location

1.13 Conditions Prior to Project Initiation

The project is a Greenfield hydro power project and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. Thus prior to project initiation, there was nothing at site.

In the absence of project activity, the continuation of current practice i.e., generation of equivalent amount of electricity would have been generated from grid connected fossil fuel dominated power plants. Thus, for project activity baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project has received all the necessary approvals for the development and commissioning of the proposed 15MW hydro power project from the respective nodal agencies and is in compliance to the local laws and regulations. The list of applicable laws & regulations and the project compliance against it are given below:

Laws & Regulation	Project Compliance
Law on Electricity (amended version) No. 03/NA, dated 20/12/2011;[2]	The project signed PPA with Electricite du Laos, a state-owned enterprise which is also an approval for electricity connection with the grid.
Environmental Protection Law (2013 Ed.) (Law No. 29/NA of 2012)[3]	Project received Environmental clearance certificate from Department of Natural Resources and Environment based on the IEE report submitted that confirms the project compliance with these laws & regulation.
Decree on Environmental Impact Assessment dated January 31, 2019[4]	
National Policy on Environmental and Social Sustainability of the Hydropower Sector in Lao PDR[5]	

[2] https://policy.asiapacificenergy.org/sites/default/files/Electricity%20Law.pdf

[3] https://policy.asiapacificenergy.org/sites/default/files/Environmental%20Protection%20Law%202013%20Revised%20Ed%20%28Lao%29.pdf

[4] https://data.laos.opendevelopmentmekong.net/en/laws_record/decree-on-environmental-impact-assessment

[5] https://policy.asiapacificenergy.org/sites/default/files/National%20Policy%20on%20the%20Environmental%20and%20Social%20Sustainability%20of%20the%20Hydropower%20Sector.pdf

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project is not registered under any other GHG programs. The project is seeking registration only in VCS program.

1.15.2 Projects Rejected by Other GHG Programs

The project is not rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

Supply Chain (Scope 3) Emissions

Have the owner(s) or retailer(s) of the impacted goods and services posted a public statement saying, "VCUs may be issued for the greenhouse gas emission reductions and removals associated with [organization name(s)] [name of good or service]" since the project's start date?

☐ Yes ☒ No

Has the project proponent posted a public statement saying, "VCUs may be issued for the greenhouse gas emission reductions and removals associated with [name of good or service] [describe the region or location, including organization name(s), where practicable]."

☐ Yes ☒ No

Have the producer(s) or retailer(s) of the impacted good or service been notified of the project and the potential risk of Scope 3 emissions double claiming via email?

☐ Yes ☒ No

As per the Corporate Value Chain (Scope 3) Accounting and Reporting Standard[6], the project activity does not involve any direct impact emissions associated with goods or services. The project activity is not related with scope 3 emissions. Hence this section is not applicable to the project activity.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

Social well-being: The project would help in generating employment opportunities during the construction and operation phases. The project activity leads to development in infrastructure in the region like development of roads and also may promote business with improved power generation.

Economic well-being: The project is a clean technology investment in the region, which would not have been taken place in the absence of the VCS benefits the project activity will also help to reduce the demand supply gap in the state.

The project activity will generate power using hydro energy-based power generation which helps to reduce GHG emissions and specific pollutants like SO_x, NO_x, and SPM associated with the conventional thermal power generation facilities.

Technological well-being: The successful operation of project activity would lead to promotion of hydro power generation and would encourage other entrepreneurs to participate in similar projects.

Environmental well-being: Hydro energy being a renewable source of energy, it reduces the dependence on fossil fuels and conserves natural resources which are on the verge of depletion. Due to its zero emission the Project activity also helps in avoiding significant amount of GHG emissions.

1.17.2 Sustainable Development Contributions Activity Monitoring

The project supplies clean electricity from the hydro power plant to the National grid hence displacing the electricity generated from grid connected fossil fuel power plants and thereby avoiding the equivalent Carbon dioxide which is a greenhouse gas. In the absence of the project activity, there won't be any avoidance of carbon dioxide.

During the construction, operation and maintenance of the hydro power plant, the project created many direct and indirect new job opportunities mostly to the local people in the project region. In the absence of the project activity, there would be no new jobs created in the project area.

As a part of regional development efforts associated with the project, PP has supported many education, health & infrastructure related needs for local people. These are funded from the revenue generated from the operation of the project activity. In the absence of project activity,

[6] https://ghgprotocol.org/sites/default/files/standards/Corporate-Value-Chain-Accounting-Reporing-Standard_041613_2.pdf

there would be no revenue generated from the project and hence the activity would have not occurred in the absence of the project activity.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions over project lifetime
1)	7	7.2 By 2030, increase substantially the share of renewable energy in the global energy mix	Implemented activities to increase	Project target to generate and feed 87,280 MWh/year to the national grid and thus expected to increase the share of renewable energy in the grid.	In the absence of the project activity 87,280 MWh/year of electricity would have been generated by the fossil fuel dominated grid. The project is expected to generate 2,182,000 MWh of electricity to the grid for its lifetime.
2)	8	8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.	Implemented activities to increase	Project creates employment to the people	Project creates new employment and generates income for the people during the project lifetime.
3)	13.0	13.2 Integrate climate change measures into national policies, strategies, and planning	Implemented activities to increase	Project target to reduce 31,453 tCO_2 emissions per year in the host country.	In the absence of the project activity 31,453 tCO_2/year emission reductions would have been generated from the fossil fuel dominated power plants in the host country. The project is expected to reduce 786,337 tCO_2 emissions in the host country for its lifetime.

1.18 Additional Information Relevant to the Project

Leakage Management

Not Applicable as no leakage emission is involved in this project.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not Applicable.

2 SAFEGUARDS

2.1 No Net Harm

The project activity involves major construction activity and primarily requires the installation of the turbines, interfacing the generators with the State Electricity Board by setting up HT transmission lines and installation of other accessories. Hydro Power project activity operations do not result in direct air pollution, noise pollution. Thus, there is no any significant impact due to implementation of project activity on air, water, soil quality and ambience are envisaged due to the project activity.

Safe-guarding Principles: The project does not involve and is not complicit in involuntary resettlement.

Relevance to the Project Activity: As per the final project design, there is no relocation involved. There is no considerable change in the water storage area. Hence, no negative impact due to the project.

Risks to Stakeholders and the Environment

There is no major risks or negative impacts associated with the operation of the project activity. Since the project is situated away from the vicinity of the inhabitants, there is no direct risk involving the inhabitants due to the project activity. The project owner ensures that safety training and hands-on training are conducted for the workers, engineers, contractors etc, involved in the project on-site daily. The grievance addressal register is made available to the local stakeholders all the time to register their grievances or complains if any due to the operation of the project activity. Hence the project owner ensures that there no impact on the stakeholders due to the project activity.

Respect for Human Rights and Equity

The project owner strictly avoids any discrimination practices while hiring people from different race, gender, ethnics, religion, marginalized groups, people with disabilities.

The project owner adheres to fair labour practices and ensure safe working conditions for their employees. This includes providing adequate training, protective equipment, and compliance with labour regulations. The project owner ensures compliance to the International Bill of Human Rights and universal instruments relating to human rights in project design and implementation. The project owner ensures that equality of opportunity and treatment of all individuals to fully develop their talents and skills according to their aspirations and preferences, and to enjoy equal access to employment as well as equal working conditions.

Property Rights

The project owner possesses the land ownership documents and ensures clear property rights boundaries. The project owner ensures there shall not encroach on private, stakeholder, or government property. Hence the project owner ensures that there no impact on the property rights due to the project activity.

Ecosystem Health

The project activity is hydro power project at Xayabouly Province, Lao PDR. The project site is predominantly secondary and disturbed forestland and patches of remnant woodland. The project site has no protected areas such as National Park, Wildlife Sanctuaries, Reserve and Protected Forest areas within the project boundary, or either, the project area is the part of any protective or eco-sensitive area. The project site and the surrounding land does not habitats for rare, threatened, or endangered species. Hence, the project has no negative impacts on the ecosystem health.

2.2 Local Stakeholder Consultation

The details of the stakeholder meetings are as follows:

Date of meeting: 21/08/2022

In accordance with Lao government Safeguard policy policies, disclosure of information to project stakeholders and the general public is necessary. The information disclosure was disseminated through Lao TV and radio to reach the general public - updates about progress on the project and public meeting announcements.

Public consultation was carried out at different levels including the Government Authorities from National, Provincial, District and village authorities and institutions, as well as international organizations. At the village level, the consultation focused on the proposed affected villages which are located in upstream, downstream and project construction sites in order to provide basic information about the project. The content of public consultation includes the provision of basic information about the project, its objectives and the expected potential project impacts and mitigation measures. The representative of each village group who attended the meeting was asked to introduce the project to the other members of each village.

Most of the inhabitants living in Houaykeng village belong to the Hmong-Mien ethnic group which represents over 65% of its total population. While those residing in Pakhoung are mostly Lao Loum or Lao ethnic groups covering 63% of the village population. They have been resettled over the last decade from approximately 3 km upstream of the existing Houaykeng village due to the government's policy of encouraging movement from isolated villages to closer public services by offering relocation incentives (land user rights) while some moved voluntarily to gain access to government services or following upgraded road alignments where land is available. However, the majority of these villages are characterized by limited development and mainly poor economy.

According to the district socio-economic report 2007-2008, it is indicated that Xayaboury district comprises 83 villages, with 12,082 households. The total population is 67,220 of which 33,919 are female. The estimation of the annual population growth is 2.9%.

There are 167 households in two villages surveyed with 181 families. The total population is 1,228 people, of whom 472 are female.

The public consultation at village level was conducted informally during the time of social economic survey in each impact village. The participants were from the different groups as in following:

- Representative of government official of the concerned districts;
- Representative of Lao women's union, at village level, Lao National Front and the Lao Youth organization at village level;
- Head of village and head of household;
- The socio-economic survey teams.

Figure 1: Public Consultation

The questions raised by the villagers are summarised in the table below:

Table.4: Stakeholder Comments

Stakeholder Comment	Explanation (Why? How?)	Remarks
Is the air in local context polluted?	Project Owner's representative confirmed that there would be no negative permanent effect to locals during the project construction and operation phase, instead there might be only minor temporary impact due to dust emission while the mitigation measure would be adopted.	The following mitigation measures followed during the construction to avoid dust. Localized sprinkling of water to supress the dust and suspended particles.
How is the air in local context before and during the construction?	The sources of air emission and fugitive dust will likely be from the drilling, crushing, and hauling associated with the construction activities. However, the dam construction is not directly linked to industrial processes, and thus the impact of emission from specific contaminants, organic chemical compounds and particulate matters will be minimized except where the work involves welding and aerosol spraying. Furthermore, the representative mentioned that a series of mitigation measure would be prepared for the local people to improve their local air.	The following mitigation measures followed during the construction to avoid dust: Localized sprinkling of water to supress the dust and suspended particles
Does this project affect the protection of biodiversity?	The project area is of predominantly secondary and disturbed forestland and patches of remnant woodland. There is no presence of primary forest nor are there any NPAs, PPAs or DPAs located within or nearby the Project area. These village conservation forests will, among other uses, provide habitats for wildlife and healthy forest ecosystem services. It is important for the project to make sure that threats to species will not be resulted from the presence of the construction workforce. At the same time, presence of the workforce should not result in increased pressures on the local harvest and use of NTFPs. For this reason, direct project impact on sensitive forest areas and resources will be minimal during the construction phase and negligible during operation.	Not Applicable

Will the implementation of the project help improve the working environment and quality of the workers?	The working environment of the workers will strictly implement as following measures: Promotion of preventive measures, e.g., use of condoms. Regular site clean-ups, draining water bodies that can harbour disease vector (mosquito). Conduct hygiene and sanitation awareness programmed. Support local initiatives on disease prevention campaigns. Siting fuel depot away from the living buildings and residential areas.	Not Applicable
Will the project provide jobs and generate incomes?	Project Owner's representative mentioned that all the construction works would be open for local construction company, and would request the company to recruit locally.	Not Applicable

Summary of the consultation meeting

Nam Houng - 1 Hydro Power project received support from all the villages around the project area. The main positive impacts of the project are as follows:

- Opportunity for employment by the Project during the construction and the operation phases;
- Opportunity to sell small goods and/or provide food for the workers at the camp;
- Benefits from an improved road-increased and all-year access;
- Availability of electricity and the opportunity to expand into small home-run businesses as well as improving everyday life; and
- Access to training and technical programs delivered by District Government (Improved access to agricultural extension services).

Post the stakeholder meeting, all the concerns raised during the meeting was properly addressed along with mitigation measures to minimise any impacts. The project proponent has a grievance mechanism. Stakeholders can anytime provide their inputs or concerns over the project through the grievance register maintained at the site.

The site manager is responsible for addressing the grievances received from stakeholders. The same is informed during the stakeholder consultation meeting. This mechanism will ensure to continue to engage with local stakeholder in an ongoing proactive manner. During the current monitoring period, no grievances received from any stakeholder.

2.3 Environmental Impact

Project Phase	Environmental and Social Impacts		Mitigation Measures
	Positive	Negative	
1. Planning		Land acquisition	All acquired land was rightly compensated according to the Laws and approval by District Authorities.
2. Construction	Creation Employment for local community including services		Training and preparing skill and unskilled labor list.
		Soil Erosion from earth, rock excavation for weir, coffer and road construction	All surplus earth and stone were reused for protection construction, new road and road renovations. Planting grass and shrub in the weak soil area was done.
		Noise Impact from rock explosion and excavation	To concentrate the explosion times, explosion was avoided at noon/nighttime, using short delay blasting technology, personal protection equipment.
		Dust pollution	Water was spraying throughout the construction activity area to keep the dust low.
		Traffic interruption	Traffic safety management, traffic warning signs.
		Impact on tree adjacent to power transmission line	Avoided tall trees during the laying of transmission line. If unavoidable, the top of trees was trimmed regularly. Workers were trained in correct techniques of tree trimming without damage to trunk or roots.
		Water pollution from weir construction	A temporary sedimentation tank with a capacity of not less than 30 m^3 was built to avoid soil discharge into the river. The existing and construction new water supply system was improved for the downstream villages.
3.Operation	Creating employment for local people, including		Training and preparing skill and unskilled labour list. Prepare tourism management promotion plan.

Project Phase	Environmental and Social Impacts		Mitigation Measures
	Positive	Negative	
	services and tourism activities, increasing local electricity supply, decreasing fire wood		
	Opportunity of water use for agriculture and aquaculture and tourism		Xayabouly District is to be promoted.
		Water flow changes under the weir, which will impact on aquatic ecologically downstream	Downstream of the power house will not be impacted by the Project
		Noise produced by machine. Domestic waste and waste water	Install soundproof doors or windows. Waste water will be treated by a small biological treatment planted underground. Some excrement will be using as organic fertilizer. Waste will collect in designated areas with anti-leakage measures (cement paved areas, awning) and then will be transferred to the garbage disposal area of the Xayabouly town that has more hygienic and safety measures.
		Waste machine oil produced by maintenance of machine and equipment.	Collect waste oil and transfer them to Xayabouly District solid waste place
		Sediment	A simple wash out facility into the structures. Sedimentation will occur at the very low load

2.4 Public Comments

No public comments were raised during the public comment period for this project.

2.5 AFOLU-Specific Safeguards

Not Applicable as the project is not a AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Title: "Small-scale methodology: Grid-connected renewable electricity generation" – AMS I.D[7]

Version: AMS I.D – Version 18.0

Type I: Energy industries (Renewable/ non-renewable sources)

Tools used along with the above methodology are mentioned below:

- Tool to calculate the emission factor for an electricity system - Version 07.0[8]
- Demonstration of additionality of small-scale project activities – Version 13.1[9]
- Investment analysis, version 12.0[10]

Methodology referred for calculation of project emissions as per AMS -I.D :

ACM0002: Grid-connected electricity generation from renewable sources, version 21.0[11]

3.2 Applicability of Methodology

The project activity involves generation of grid connected electricity from hydro energy. The project activity has a proposed capacity of 15 MW which will qualify for a small scale CDM project activity under Type-I of the small-scale methodologies. The project status is corresponding to the methodology AMS I.D version 18.0 and applicability of methodology is discussed below.

[7] https://cdm.unfccc.int/UserManagement/FileStorage/2P7FS6ZQAR84LG3NMKYUH50WI9ODBC

[8] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf

[9] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf

[10] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf

[11] https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL01OU2RYC6N3HASIXV7K84QBG9

Small-scale methodology: G*rid connected renewable electricity generation – AMS I.D* version 18.0	
Applicability Criteria	**Applicability Status**
This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	The project activity involves setting up of a renewable energy (hydro) generation plant that exports electricity to the fossil fuel dominated electricity grid. Thus, the project meets this applicability conditions "a"
Illustration of respective situations under which each of the methodology (i.e., AMS-I. D: Grid connected renewable electricity generation", AMS-I.F: Renewable electricity generation for captive use and mini-grid" and AMS-I.A: Electricity generation by the user) applies is included in Table 2	According to the point 1 of the Table 2 in the methodology – "Project supplies electricity to a national/ regional grid" is applicable under AMS I.D. As the project activity supplies the electricity to Lao PDR Grid system grid which is a regional grid, the methodology AMS-I.D. is applicable.
This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s); or (e) Involve a replacement of (an) existing plant(s)	The Project activity involves the installation of new hydro power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity. Thus, Project activity is a Greenfield plant and satisfies this applicability condition (a).
Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in existing reservoir, with no change in the volume of the reservoir; or (b) The project activity is implemented in existing reservoir, where the volume of the reservoir(s) is increased and the power density as per definitions given in the project emissions section, is greater than 4 W/m^2. (c) The project activity results in new reservoirs and the power density of the power	The project activity is construction of a new hydro power plant (i.e., Greenfield project) which results in a new reservoir and the power density as per definitions given in the project emissions section, is greater than 4 W/m^2. Hence, meeting with this criterion.

plant, as per definitions given in the project emissions section, is greater than 4 W/m^2	
If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The rated capacity of the project activity is 15 MW with no provision of Co-firing fossil fuel. Hence, meeting with this criterion.
Combined heat and power (co-generation) systems are not eligible under this category.	This is not relevant to the project activity as the project involves only hydro power generating units.
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	There is no other existing renewable energy power generation facility at the project site. Therefore, this criterion is not applicable.
In the case of retrofit or replacement, to qualify as a small-scale project, the total output of the retrofitted or replacement power plant/unit shall not exceed the limit of 15 MW.	The project activity is a new installation, it does not involve any retrofit measures nor any replacement and hence is not applicable for the project activity.
In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored	This project is a hydro power project activity and hence this criterion is not applicable.
In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	This project is a hydro power project activity and hence this criterion is not applicable.

In addition, the above applicability conditions, the applicability conditions of tool referred in the methodology AMS I.D, version 18.0 has been referred here under:	
Tool 07: Tool to calculate the emission factor for an electricity system version 7.0	
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes of grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g.: demand-side energy efficiency projects).	The project activity is a greenfield hydro power generation plant and hence, according to the applied methodology, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in "TOOL07: Tool to calculate the emission factor for an electricity system".
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, the conditions specified in "Appendix 2: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Since the project activity is grid connected, this condition is applicable and the emission factor has been calculated accordingly.
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in Lao PDR, a non-Annex I country. Therefore, this tool is applicable for the project activity.
Under this tool, the value applied to the CO_2 emission factor of bio fuels is zero.	The project activity is a grid connected hydro power project and therefore, this criterion is not applicable for the project activity
Tool 21: Demonstration of additionality of a small-scale project activity – version 13.1	

The use of the methodological tool "Demonstration of additionality of small-scale project activities" is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board	The project proponent is not proposing any new methodology and has applied this tool for the demonstration additionality in compliance with the tool. Refer to section 3.5, for the detailed applicability of this tool and additionality assessment. Hence this tool is applicable.
Tool 27: Investment Analysis – version 12.0	
This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", the guidelines "Non-binding best practice examples to demonstrate additionality for SSC project activities", or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	PP used "demonstration of additionality of small-scale project activities" and "non-binding best practice examples to demonstrate additionality for SSC project activities" to demonstrate additionality of the project. Hence, Tool 27: Investment analysis is applicable for the project activity.
In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied methodology AMS I.D does not contain requirement for the investment analysis. Hence this criterion is not applicable

3.3 Project Boundary

As per applicable methodology AMS-I.D. Version 18, "The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the projectpower plant is connected to."

Thus, the project boundary includes the Hydro Turbine Generators and the Lao grid system.

The project does not involve any other emissions sources not foreseen by the methodologies. The greenhouse gases and emission sources included in or excluded from the project boundary are shown intable below.

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

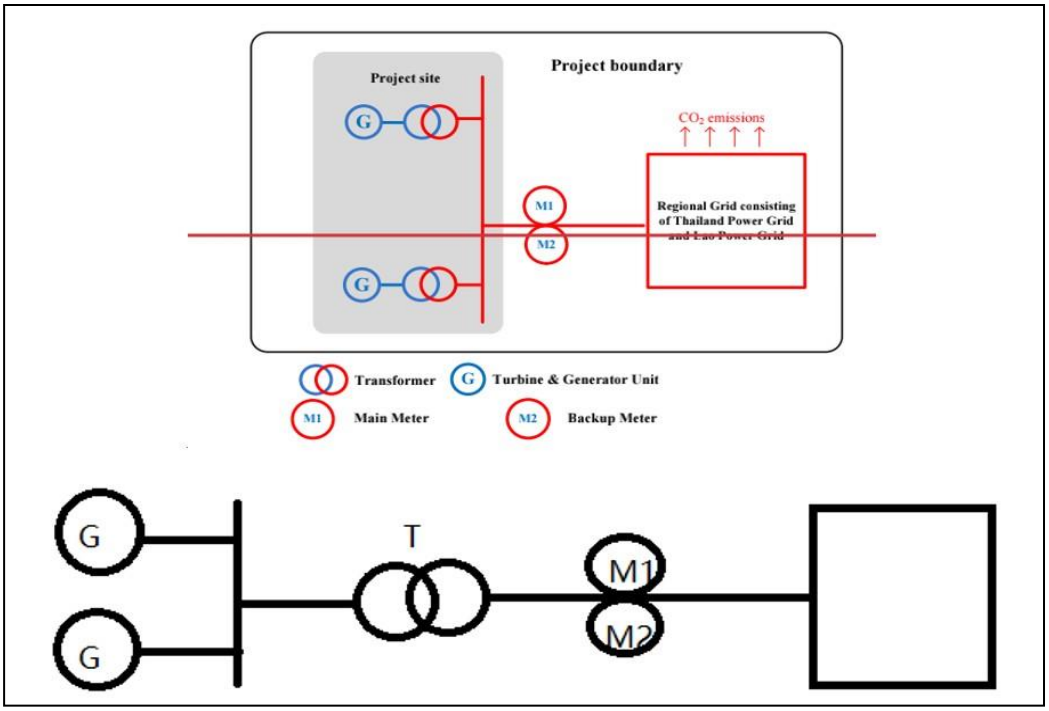

Fig. 2 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	CO_2 emissions from electricity generation in fossil fuel power plants that are displaced due to the project activity	CO_2	Yes	Main Emission Source
		CH_4	No	Minor Emission Source
		N_2O	No	Minor Emission Source
Project	Greenfield hydro power project activity Emission of CH_4 from the reservoir	CO_2	No	Project activity does not emit CO_2
		CH_4	No	Project activity does not emit CH_4
		N_2O	No	Project activity does not emit N_2O

3.4 Baseline Scenario

As per the approved consolidated Methodology AMS I.D (Version 18.0) para 19: "If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system".

The project activity involves setting up of hydro project to harness the power of hydro energy to produce electricity and supply to the grid. In the absence of the project activity, the equivalent amount of power would have been supplied by the national grid, which is fed mainly by fossil fuel fired plants.

The project is a hydropower project. As described in Section B.3 of the PDD, the project electricity system is a regional grid consisting of Thailand Power Grid and the Lao Power Grid. In the absence of the project, the local was/will be supplied by the above-mentioned regional grid. Thus, the baseline scenario of the project is continuation of the present situation, i.e., electricity supplied from the regional power grid.

The combined margin emission factor of the grid used for the project activity is as follows:

Parameter	Value	Nomenclature
$EF_{grid,CM,y}$	0.4504 tCO_2/MWh	Combined margin CO_2 emission factor for the project electricity system in year y
$EF_{grid,OM,y}$	0.4504 tCO_2/MWh	Operating margin CO_2 emission factor for the project electricity system in year y
$EF_{grid,BM,y}$	0 tCO_2/MWh	Build margin CO_2 emission factor for the project electricity system in year y

3.5 Additionality

As per the Law on Electricity (amended version) No. 03/NA, dated 20/12/2011 implementation of the renewable energy project is not mandated and are entirely a voluntary action. The project activity does not fulfil the criteria of positive list as provided in CDM Tool 32: “Methodological Tool – Positive List of Technologies” and hence additionality of the project activity is demonstrated through a project specific additionality test.

For the demonstration and assessment of additionality, “Tool 21: Demonstration of additionality for small-scale project activities”, Version 13.1 has been referred which further refers to guidelines of “Non-binding best practice examples to demonstrate additionality for SSC project activities”. As per the para 10 of the tool 21, Project owner shall provide an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

Investment Barrier: a financially more viable alternative to the project activity would have led to higher emissions
Technological barrier: a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity as so would have led to higher emissions:
Barrier due to prevailing practice: or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions.

Other barriers: without the project activity, for another specific reason identified by the project participant, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emissions would have been higher.

The project owner has selected Investment barrier to demonstrate in a conservative and transparent manner that the project activity is financially unattractive.
To conduct the investment analysis, Methodological tool 27: Investment analysis, version 12, EB 116 Annex 2[12] has been referred.

Considering the fact that the alternative to the project is the supply of electricity from the grid & the choice of the developer is to invest or not to invest, benchmark analysis has been considered appropriate for demonstration of additionality, which is in conformity with "Tool 27: Investment Analysis, version 12.0".

For financial analysis of the project, "Tool 21: Demonstration of additionality for small-scale project activities", Version 13.1 has been applied which further refers to the guidelines of "Non-binding best practice examples to demonstrate additionality for SSC project activities" which states the following under investment barrier:

"Application of investment comparison analysis using a relevant financial indicator, application of a benchmark analysis or a simple cost analysis".

The Project activity envisages to export the power to National grid and the revenues from the sale of electricity would be generated in accordance with the terms and tariffs established in the Power Purchase Agreement (PPA). Thus, simple cost analysis cannot be used as the analysis method as the sale of the units of generated electricity shall result in a revenue stream during the operations of the Project activity.

In the absence of the project activity grid electricity would have been the obvious choice for the Project which requires no investment. Hence, investment comparison analysis is also not appropriate for the project activity.

However, after eliminating simple cost analysis and investment comparison analysis, the use of Benchmark analysis is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be employed for analysis of the said project.

Benchmark analysis

As per para 15 of Tool 27: Investment analysis, Version 12 states that Required/expected returns on equity are appropriate benchmarks for equity IRR. The project participant has chosen

[12] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf

benchmark analysis to demonstrate the additionality of the project. The project is promoted by private limited company and hence the return on equity and the risks associated with the investments for their shareholder is of primary concern. Hence, in order to analyze the financial viability of the project activity, the prime financial indicator that has been used is the post-tax equity IRR of the project activity.

Selection of Appropriate Benchmark

The benchmark has been considered in accordance with paragraph 19 of "Tool 27: Investment Analysis, version 12.0, "The values in the table in the Appendix may also be used, as a simple default option".

As the project activity generates power utilizing hydro energy, Group 1 as per para 5a of Appendix of EB 116 Annex 2 has been identified as a suitable category.

- The investment analysis has been carried out in Nominal terms. Accordingly, default value as given in table under the Appendix, Tool 27 has been adjusted by adding suitable forecasted inflation rate taken from IMF world Economic database as no inflation forecast or inflation target published by Lao PDR
- Average forecasted inflation rate for the host country (Lao PDR) published by the IMF (International Monetary Fund) World Economic Outlook for the next five years has been considered.

Hence applicable inflation rate has been chosen from the IMF databases accordingly for the estimation of resultant benchmark.

As per para 19 of investment analysis, the cost of equity is determined by selecting the values provided in the Appendix. i.e., Default values for cost of equity (expected return on equity) is presented below:

Default Value Benchmark:

The cost of equity is determined by selecting the values provided in the table of the Appendix, i.e., Default values for cost of equity (expected return on equity) in the 'Tool 27: Investment analysis', version 12.0.

The Required return on equity (benchmark) was computed in the following manner:

Nominal Benchmark[13] = {(1+Real Benchmark) * (1+Inflation rate)}-1

Where:

Default value for Real Benchmark = 20.68% [14] (CDM Tool 27: Investment Analysis, version 12)

Inflation Rate forecast for by (International Monetary Fund) World Economic Outlook database for Lao PDR.

Benchmark Estimation:

The selection of the benchmark estimation is based on the Cost of Equity has been considered using the "Tool 27: Investment analysis", version 12.0 available at the time of initial submission.

[13] As per Pg.320 of Corporate Finance, Second Edition of Aswath Damodaran

[14] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf

Table under Appendix in EB116, Annex 2 specifies default value of expected return on equity in real terms for Energy Industries in Lao PDR = **20.68%**

Thus, minimum cost of equity considered for calculation of Benchmark = 20.68%
5-year inflation Forecast for Lao PDR as IMF World Economic Outlook available at the time of investment decision and corresponding benchmark values is

Inflation forecast for 5 years	Source
2.90%	World Economic Outlook database: April 2016[15]

Corresponding benchmark values applicable at the time of investment decision time (i.e. 24-September-2016, which is the signing date of the EPC contract):
Nominal Benchmark = {(1+20.68%) * (1+2.90%)} - 1 = 24.18%

Calculation and comparison of financial indicators
The period considered for Post Tax Equity IRR calculations is 30 years, which corresponds to the operational lifetime of the project activity.

Depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, is added back to net profits for the purpose of calculating the financial indicator.

Parameters for IRR calculation

Particulars	Value	Unit	Source/Remarks
Capacity of the project	15	MW	Feasibility Report dated June 2016 Pg No 195
Annual Net generation	87.280	GWh	Feasibility Report dated June 2016 Pg No 195
Project cost	45186.20	k USD	Feasibility Report dated June 2016 Pg No 195
Debt	80%	%	Feasibility Report dated June 2016
Equity	20%	%	Feasibility Report dated June 2016
Debt	36148.96	K USD	Calculated
Equity	9037.24	K USD	Calculated
Interest rate	10.00%	%	Feasibility Report dated Jan 2016 Pg 198
Debt Repayment tenure	12	years	Feasibility Report dated Jan 2016 Pg 198
Moratorium	0	year	Feasibility Report dated Jan 2016 Pg 198
Royalty (% of gross revenue)	5.00%		Feasibility Report dated Jan 2016 Pg 198
Insurance (% of fixed asset)	75.31033	k USD	Feasibility Report dated Jan 2016 Pg 198
Transmission line maintenance	6.632	k USD	Feasibility Report dated Jan 2016 Pg 198
O&M Labour	10.5	k USD	Feasibility Report dated Jan 2016 Pg 198
Tariff	0.065	USD/kWh	PPA & Feasibility Report dated Jan 2016 Pg 198
Depreciation Rate (Book)	4.00%	%	Feasibility Report dated Jan 2016 Pg 157

[15] https://www.imf.org/en/Publications/WEO/weo-database/2016/April/weo-report?c=544,&s=PCPI,PCPIPCH,&sy=2017&ey=2021&ssm=0&scsm=1&scc=0&ssd=1&ssc=0&sic=0&sort=country&ds=.&br=1

Salvage Value	10%	%	Assumption
Income tax rate for year 1 to year 10	10%	%	Feasibility Report dated Jan 2016 Pg 200
Income tax rate for year 11 to year 30	24%	%	Feasibility Report dated Jan 2016 Pg 200

IRR for total project capacity

Project proponent also calculated the Post tax Equity IRR by integrating the total cash flows of all three projects to check the viability of the project against the benchmark value. Still, it is evident that the combined project is also not financially viable as the equity IRR is still less below the benchmark

Post Tax Equity IRR for the project activities against the benchmark values are shown in table below. Thus, it is evident that the project is not financially attractive as the equity IRR is less below the benchmark value.

Post Tax Equity IRR	9.92%
Benchmark	24.18%

Sensitivity Analysis

The robustness of the conclusion drawn above, namely that the project is not financially attractive, has been tested by subjecting critical assumptions to reasonable variation. As required by Annex 2 of EB 116, only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. Project owner has identified the total revenue from the project activity is dependent on the Tariff, Plant Load Factor, Project Cost and O&M Costs constitute more than 20% of the project costs. These factors have been subjected to a 10% variation on either side and the results of the sensitivity analysis indicate that even after applying such variation the EIRR does not cross the benchmark.

Variation %	**-10%**	**Normal**	**10%**	**Variation required to reach benchmark**	**Value to breach benchmark**
Tariff	7.83%	9.92%	12.09%	59.50%	0.104 USD/kWh
Generation	7.83%	9.92%	12.09%	59.50%	139.212 GWh
Project Cost	12.30%	9.92%	8.05%	-37.70%	28151.00 k USD
O&M Cost	10.07%	9.92%	9.77%	-852.30%	-2829.42 k UDS

Conclusion

An analysis has been done to identify the percentage variation at which the financial indicators will equal/breach the benchmark and the probability of its occurrence. Based on sensitivity analysis it can be concluded that the project activity is additional with reasonable variation in values and is not likely to reach the benchmark value. The occurrence of these events is unlikely for the following reasons:

a) **Tariff:** The Tariff rate of electricity used for investment analysis i.e., 0.065 USD/kWh is sourced from the PPA. Furthermore, the project will breach the benchmark value at a tariff variation of 59.50%. Hence, 59.50% increase in tariff is unlikely as the PPA is already signed.

b) **Generation:** The annual net generation considered is based on the feasibility report which is 58.090 GWh and the IRR breach the benchmark value at a Generation of more than 59.50% increase from the base value. Hence the generation considered in for the financial calculation is conservative.

c) **Project Cost:** The project cost considered for investment analysis i.e. 45186.20 k USD. The cost is sourced from feasibility report. A variation of -37.70% is required for IRR to breach benchmark which is not possible as the project is already commissioned. The actual cost incurred in commissioning of the project is same as the estimated cost as per the investment certificate and hence the IRR doesn't breach the benchmark at actual project costs. Hence reduction of 37.70% is unlikely to happen and the analysis holds good.

d) **O&M Costs:** The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is subject to escalation (as evidence by the O&M agreement) and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely. Even after the application of actual value, the equity IRR is much below the benchmark value. Hence the analysis holds good.

3.6 Methodology Deviations

There is no methodology deviation.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline emissions include only CO_2 emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$\mathbf{BE_y = EG_{PJ,y} \times EF_{grid,y}}$ (Equation 1)

Where,

BE_y = Baseline emissions in year y (t CO_2/yr)

$EG_{PJ,y}$= Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO_2/MWh)

If the project activity is the installation of a greenfield power plant, then:

$\mathbf{EG_{PJ,y} = EG_{PJ,facility,y}}$ (Equation 2)

Where,

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

Calculate the $EF_{grid,y}$

The emission factor should be calculated in a transparent and conservative manner according to the procedures prescribed in the "Tool to calculate the emission factor for an electricity system" (Version 07.0).

The data used for calculation are from an official source (where available) and publicly available. The calculation processes are as follows:

STEP1: Identify the relevant electricity system;
STEP2: Choose whether to include off-grid power plants in the project electricity system (optional);
STEP3: Select a method to determine the operating margin (OM);
STEP4: Calculate the operating margin emission factor according to the selected method;
STEP5: Calculate the build margin (BM) emission factor;
STEP6: Calculate the combined margin (CM) emissions factor.

STEP 1: Identify the relevant electricity system

As descripted in Section B.3., there are no transmission constraints between Lao and Thailand, the project electricity system is a regional grid consisting of Thailand Power Grid and Lao Power Grid. And as electricity imported from Malaysia, China and Vietnam Power Grid10, these three Power Grids are considered as connected electricity system.

STEP 2: Choose whether to include off-grid power plants in the project electricity system (optional)

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), there are two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.
Option II: Both grid power plants and off-grid power plants are included in the calculation.

Option I is chosen for emission factor calculation.

STEP 3: Select a method to determine the operating margin (OM)

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), there are four methods for calculating the $EF_{grid,OM,y}$:

a) Simple OM, or
b) Simple adjusted OM, or
c) Dispatch data Analysis OM, or
d) Average OM

The method (d), average OM, is selected.

$EF_{grid,OM\text{-}ave,y}$ is calculated using ex ante option: a 3-year generation-weighted average in 2020, 2019, 2018 without requirement to monitor and recalculate the emissions factor during the crediting period.

STEP 4: Calculate the operating margin emission factor according to the selected method

The average OM emission factor is calculated as the average emission rate of all power plants serving the grid, using the methodological guidance as described under Step 4 in the “Tool to calculate the emission factor for an electricity system” (Version 07.0) for the simple OM, but also including the low-cost / must-run power plants in all equations.

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), there are two options based on different data for calculating average OM:

Option A: Based on the net electricity generation and a CO_2 emission factor of each power unit; Or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

For the project, the necessary data for Option A is not available, so Option B was used.

Under this option, the average OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OM\text{-}ave,y}=\frac{\sum_i (FC_{i,y}\times NCV_{i,y}\times EF_{CO2,i,y})}{EG_y}$$

$EF_{grid,OM\text{-}ave,y\ FCi,\ y}$ = Average operating margin CO_2 emission factor in year y (tCO_2/MWh)
$FC_{i,y}$ = Amount of fossil fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$ = Net calorific value (energy content) of fossil fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$ = CO_2 emission factor of fossil fuel type i in year y (tCO_2/GJ)
EG_y = Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)
I = All fossil fuel types combusted in power sources in project electricity system in year y
y = The data available in the most recent 3 years

According to the “Tool to calculate the emission factor for an electricity system” (Version 07.0), electricity imports from the connected electricity systems $EG_{import,y}$ are included in the EG_y

The calculation value of average OM emission factor is 0.5542 tCO_2/MWh and the detailed calculating procedures please refer to ER calculation sheet.

Step 5. Calculate the build margin (BM) emission factor

To calculate the build margin (BM) emission factor, the data for determine the sample group of power units m about the most recently units in the electricity system is needed. However, as an international project system, it’s difficult to obtain the information for all the units in both Lao and Thailand (power generation data, commissioning date, and the fuel consumption). The data requirements for the application for calculate the build margin (BM) emission factor cannot be met.

As the Simplified CM is adopted in the step 6, the weighting of build margin emissions factor is 0.

STEP 6: Calculate the combined margin (CM) emissions factor

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), the calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

a) Weighted average CM; or
b) Simplified CM.

According to “Tool to calculate the emission factor for an electricity system” Version 07.0, the simplified CM can be used if:

a) The project activity is located in: (i) a Least Developed Country (LDC); or in (ii) a country with less than 10 registered CDM projects at the starting date of validation; or (iii) a Small Island Developing States (SIDS); and

b) The data requirements for the application of Step 5 above cannot be met.

The project located in Lao, which is a Least Developed Country, therefore the criteria (a) is met; As mentioned in step 5, the data requirements for the application for calculate the build margin (BM) emission factor is not available, therefore the criteria (b) is also met.

The Simplified CM method is calculated as follow:

$EF_{grid,CM,y} = w_{OM} \times EF_{grid,OM,y} + w_{BM} \times EF_{grid,BM,y}$

Where:

w_{OM} = Weighting of operating margin emission factor (%)

w_{BM}= Weighting of build margin emission factor (%).

The weighs w_{OM} and w_{BM}, for simplified CM by default, are w_{OM}=1 and w_{BM}= 0.

Hence calculation value of CM emission factor is 0.4504 tCO_2/MWh

Hence the baseline emission factor is calculated as below:

BE_y= 87,280 × 0.4504 = 39,309 tCO_2 (Rounded down value)

4.2 Project Emissions

The project activity is a hydro power project activity and as per the para 39 of the applied methodology the Emissions from water reservoirs of hydro power plants shall be determined considered following the procedure described in the most recent version of “ACM0002: Grid-connected electricity generation from renewable sources”. As per ACM0002: Grid-connected electricity generation from renewable sources V.21.0”, paragraph 35

$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$ (Equation 1)

For this project activity, the emissions from water reservoirs of hydro power plants ($PE_{HP,y}$) are applicable. The power density (PD) of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD = Power density of the project activity (W/m^2)

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W) (ie, 15,000,000 W)

Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} = Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m^2) (ie, as per the feasibility report it is 1520000 m^2)

A_{BL} = Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m^2). For new reservoirs, this value is zero

PD = (15,000,000-0)/(1,520,000-0) = 9.87 W/m^2

As per ACM002, version 21.0 methodology, If the power density of the project activity is greater than 4 W/m2 and less than or equal to 10 W/m^2, $PE_{HP,y}$ is calculated

$$PE_{HP,y} = \frac{EF_{Res} \times TEG_y}{1000}$$

Where:

$PE_{HP,y}$ = Project emissions from water reservoirs (t CO_{2e}/yr)

EF_{Res} = Default emission factor for emissions from reservoirs of hydro power plants (kg CO_{2e}/MWh) -As per ACM0002, EF_{Res} table in page number 13, it is 90 kgCO_2e/MWh

TEG_y = Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y (MWh) – as per FSR, it is 87,280 MWh per year

Hence, PE_y = (90x87,280)/1000 = 7,856 t CO_{2e}/yr (Rounded up value)

4.3 Leakage

No leakage emissions are considered.

4.4 Net GHG Emission Reductions and Removals

The ex-ante emission reductions (ER_y) for the project activity are calculated as follows

$$ER_y = BE_y - PE_y - LE_y$$

Where,
ER_y = Emission Reduction in year y (tCO_2)
BE_y = Baseline emission in year y (tCO_2)
PE_y = Project emissions in year y (tCO_2)
LE_y = Leakage Emissions in year y (tCO_2)

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
01-September-2023 to 31-December-2023	13,139	2,626	0	10,513
01-January-2024 to 31-December-2024	39,309	7,855	0	31,453
01-January-2025 to 31-December-2025	39,309	7,855	0	31,453
01-January-2026 to 31-December-2026	39,309	7,855	0	31,453
01-January-2027 to 31-December-2027	39,309	7,855	0	31,453
01-January-2028 to 31-December-2028	39,309	7,855	0	31,453
01-January-2029 to 31-December-2029	39,309	7,855	0	31,453
01-January-2030 to 31-August-2030	26,170	5,230	0	20,940
Total	275,161	54,986	0	220,174
Total number of crediting years	7			
Annual average over the crediting period	39,309	7,855	0	31,453

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO_2/MWh
Description	Operating Margin CO2 emission factor in year y
Source of data	Calculated

Value applied	0.4504
Justification of choice of data or description of measurement methods and procedures applied	Calculated as per "Tool to calculate the emission factor for an electricity system, version 07" as 3-year generation weighted average using data for the years 2018, 2019 & 2020
Purpose of Data	Baseline emission calculation
Comments	This parameter is fixed ex-ante for the entire crediting period

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO_2/MWh
Description	Built Margin CO2 emission factor in year y
Source of data	Calculated
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	Calculated as per "Tool to calculate the emission factor for an electricity system, version 07"
Purpose of Data	Baseline emission calculation
Comments	This parameter is fixed ex-ante for the entire crediting period

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2/MWh
Description	Combined margin CO_2 emission factor for the project electricity system in year y
Source of data	Calculated
Value applied	0.4504
Justification of choice of data or description of measurement methods and procedures applied	The combined margin emissions factor is calculated as follows: $EF_{grid,CM,y}= EF_{grid,OM,y}* WOM + EF_{grid,BM,y}*WBM$ Where: $EF_{grid,BM,y}$= Build margin CO_2 emission factor in year y (tCO_2/MWh)

	$EF_{grid,OM,y}$= Operating margin CO_2 emission factor in year y (tCO_2/MWh) WOM = Weighting of operating margin emissions factor (%) = 100% (for simplified CM) WBM= Weighting of build margin emissions factor (%) = 0% (for simplified CM)
Purpose of Data	Baseline emission calculation
Comments	This parameter is fixed ex-ante for the entire crediting period

Data / Parameter	EF_{Res}
Data unit	$kgCO_2e/MWh$
Description	Default emission factor for emissions from reservoirs
Source of data	ACM0002: Grid-connected electricity generation from renewable sources V.21.0
Value applied	90 $kgCO_2e/MWh$
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of Data	Power density calculation
Comments	This parameter is fixed ex-ante for the entire crediting period

Data / Parameter	Cap_{BL}
Data unit	W
Description	Installed capacity of the hydro power plant before the implementation of the project activity. For new hydro power plants, this value is zero
Source of data	ACM0002: Grid-connected electricity generation from renewable sources V.21.0
Value applied	0
Justification of choice of data or description of	Not Applicable

measurement methods and procedures applied	
Purpose of Data	Power density calculation
Comments	This parameter is fixed ex-ante for the entire crediting period

Data / Parameter	A_{BL}
Data unit	m^2
Description	Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m2). For new reservoirs, this value is zero
Source of data	Project site
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	Since this is project involves construction of new reservoirs, it is considered as zero as per methodology.
Purpose of Data	*Project Emission calculation*
Comments	This parameter is fixed ex-ante for the entire crediting period

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh/y
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y in MWh
Source of data	Electricity meter readings
Description of measurement methods and procedures to be applied	The difference of final value of export and import is used for monthly values of net electricity supplied to the grid by the project activity and same value will be considered for ER calculations.
Frequency of monitoring/recording	Continuous and monthly measurement

Value applied	87,280 MWh
Monitoring equipment	The electricity exported / supplied by the plant to pooling substation and further to substation. This meter also measures electricity imported by the plant from the grid. Energy meter with an accuracy: class 1.0 for kWh, class 2.0 for kvarh Energy meter make – DTSD341 Serial Number: 1. Main – 250318494, Backup – 250318495 2. Main – 250318493, Backup – 219014741
QA/QC procedures to be applied	The frequency of calibration of energy meters is once in 5 years. The monthly electricity supplied/exported by the project activity in the JMR report is cross checked with the monthly invoices of sale. In the absence or delay in the meter calibration appropriate Guidelines will be applied to confirm the conservativeness of metering. The metering arrangement, accuracy class of meters, calibration frequency is under control of EDL and PP does not have any control on it.
Purpose of data	Calculation of baseline emissions
Calculation method	Net electricity supplied to the grid by the project plant in a given month = Export(kWh)– Import (kWh)
Comments	Data will be archived in paper & electronic form for two years after the end of crediting period or of the last issuance of VERs for this project activity, whichever occurs later.

Data / Parameter	TEGy
Data unit	MWh/year
Description	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y
Source of data	Project Activity Site
Description of measurement methods and procedures to be applied	Electricity meters

Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	87,280
Monitoring equipment	Electricity meters
QA/QC procedures to be applied	NA
Purpose of data	Calculation of *project* emissions
Calculation method	NA
Comments	NA

Data / Parameter	Cap_{PJ}
Data unit	W
Description	Installed capacity of the hydro power plant after the implementation of the project activity
Source of data	Project Site
Description of measurement methods and procedures to be applied	Determine the installed capacity based on manufacturer’s specifications or commissioning data or recognized standards
Frequency of monitoring/recording	Once at the beginning of each crediting period
Value applied	15000000
Monitoring equipment	NA
QA/QC procedures to be applied	NA
Purpose of data	Calculation of project emissions
Calculation method	NA
Comments	NA

Data / Parameter	A_{PJ}
Data unit	m^2
Description	Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full
Source of data	Project Site
Description of measurement methods and procedures to be applied	Measured from topographical surveys, maps, satellite pictures, etc
Frequency of monitoring/recording	Once at the beginning of each crediting period
Value applied	1520000
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of *project* emissions
Calculation method	-
Comments	-

5.3 Monitoring Plan

The monitoring plan is developed to establish suitable data collection method for measurement & collection of data and maintenance of records according to the monitoring methodology of AMS I.D Version 18.0. The monitoring plan is project specific for which, the project performance with all relevant criteria will be monitored. Proper training will be provided to concerned personnel for operation purpose.

The purpose of monitoring is to calculate and monitor GHG emission reduction by the project activity. The monitoring plan, which is implemented by the project participant describes about the following aspects:

- Overall project management
- Monitoring Plan
- Emergency Preparedness
- Training on Monitoring & Archiving of Data and Internal Audit Procedures
- Quality Assurance and Quality Control (QA/QC) procedures

The project owner may seek support from the consultant to ensure quality of the collected data and monitoring procedures to be in line with the VCS requirements.

Overall project management team

The project owner organizes a separate team to be responsible for data collection, supervision and witness the whole process of data measuring and recording. The technical manager will be appointed to take full responsibility for the overall monitoring of the project. The monitoring and measurement will be carried out by designated technical manager. The technical department will be responsible for carrying out the internal auditing and AQ/AC. All the values from generation record will be checked with the invoices for consistency. The operation and management structure are shown below:

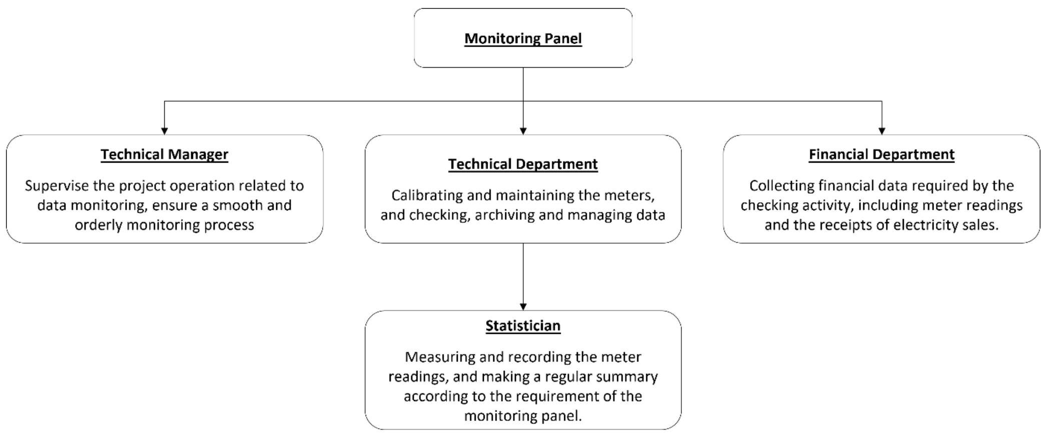

Fig. 3 Structure of the monitoring team

Monitoring plan

The meter(s) will be installed at the project site, to monitor the input/output electricity at the grid side. The meter(s) will be installed in accordance with relevant national or international standard. Before the operation of the project, the metering equipment(s) will be clarified and examined by the project owner and the power grid company according to the above regulation.

Personal Training:

The project employs qualified and experienced persons for plant operation. The training period shall be for three months, as this would be adequate and necessary to ensure proper imparting of the objective. The training course will be thoroughly and meticulously designed, highlighting the objectives, salient features, operational aspects and trouble shooting.

Procedure for handling of identified non-conformities

- The non-conformities are identified through routine inspections, monitoring systems, audits, or incident reports.

- The details of the non-conformity, including the date, time, location, and description of the event will be recorded.
- The plant operator will report the event immediately to the supervisor. Following, an initial assessment will be conducted to determine the severity and potential impact of the non-conformity on safety, operations, and compliance.
- This include stopping the process, isolating equipment, or implementing temporary fixes. Once the non-conformities are resolved, preventive actions will be implemented to avoid similar non-conformities in the future.
- This includes revising procedures, enhancing training programs, or upgrading equipment.
- The monitoring plan will be followed to continuously check for potential non-conformities and preventive measures will be reviewed periodically.

Emergency preparedness:

In case of any unforeseen event that is not covered under this monitoring plan, staff of the operation division will immediately inform the chief of the operation division. The chief of the operation division is then responsible to ensure that the cause for the unforeseen event is detected, the event is remedied and for the period of time in which the unforeseen event has occurred uncertainty in data gathered is limited as much as possible.

Data Collection:

The specific steps for data collection and reporting are listed below:

a) During the crediting period, both the EDL and the project owner will record the values displayed by the main meter.
b) Simultaneously to step a), the project owner will record the values displayed by the backup meters.
c) The meters will be calibrated according to the relevant regulation and request of EDL
d) The main meter's readings will be cross-checked with record document confirmed by EDL.
e) The project owner and the EDL will record both output and input power readings from the main meter. These data will be used to calculate the amount of net electricity delivered to the grid.
f) The project owner will be responsible of providing copies of record document confirmed by EDL to the DOE for verification.

   If the reading of the main meter in a certain month is inaccurate and beyond the allowable error or the meter doesn't work normally, the grid-connected power generation shall be determined by following measures:

   a) Read the data of the backup meters.
   b) If the backup meter's data is not so accurate as to be accepted, or the practice is not standardized, the project owner and the grid corporation should jointly make a reasonable and conservative estimation method which can be supported by sufficient evidence and proved to be reasonable and conservative when verified by DOE.

c) If the project owner and the EDL don't agree on an estimated method, arbitration will be conducted according the procedures set by the agreement to work out an estimation method.

Internal Auditing:
Project owner will conduct the internal auditing by cross checking the data from the invoices and data logbooks for ensuring the quality and consistency of data measured and monitored for the purpose of emission reduction calculations.

Quality Assurance and Quality Control (QA/QC) procedures

Calibration of Measuring Equipment's:

The reliability of the monitoring system depends on the accuracy of the measuring instrument and the quality of the relevant equipment. Thus, the meters will be regularly calibrated as per the frequency specified by the manufacturer. To assist in future verifications, the project owner will preserve the calibration records, along with the data files of project monitoring. Error check routines will be established on site and at the point of data storage to detect data measuring as well as malfunctions.

In the case of malfunction of the meters, the meter supplier will provide technical support to engage the problem promptly and emission reductions during the corresponding period will be calculated conservatively.

Monitoring Report

Physical document such as the plant electrical wiring diagram will be gathered with this monitoring plan in a single place. In order to facilitate auditors" access to project documents, the project materials and monitoring results will be indexed. All paper-based information will be stored by the technical department of the project owner and all the material will have a copy for backup. All data, including calibration records, will be kept until 2 years after the end of the total crediting period.

During the crediting period, at the end of each year, the monitoring officer shall produce a monitoring report covering the past monitoring period. The report shall be transmitted to the General Manager who will check the data and issue a final monitoring report in the name of the project participants. Once the final report is issued, it will be submitted to the DOE for verification.